

PROFIT ANALYSIS USING REGRESSION ANALYSIS

Internship Project – PRDA-01

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Problem Statement

- The objective of this project is to analyze how **R&D Spend**, **Marketing Spend**, and **Administration Spend** influence startup profits using **Regression Analysis**. The project also aims to predict future profit values and provide business recommendations based on data insights.
- The goal is to identify key profit drivers and support data-driven business decisions.

Dataset Overview

The dataset contains information from 50 startups located in New York, California, and Florida.

Features:

- **R&D Spend**
- Administration
- Marketing Spend
- State
- Profit

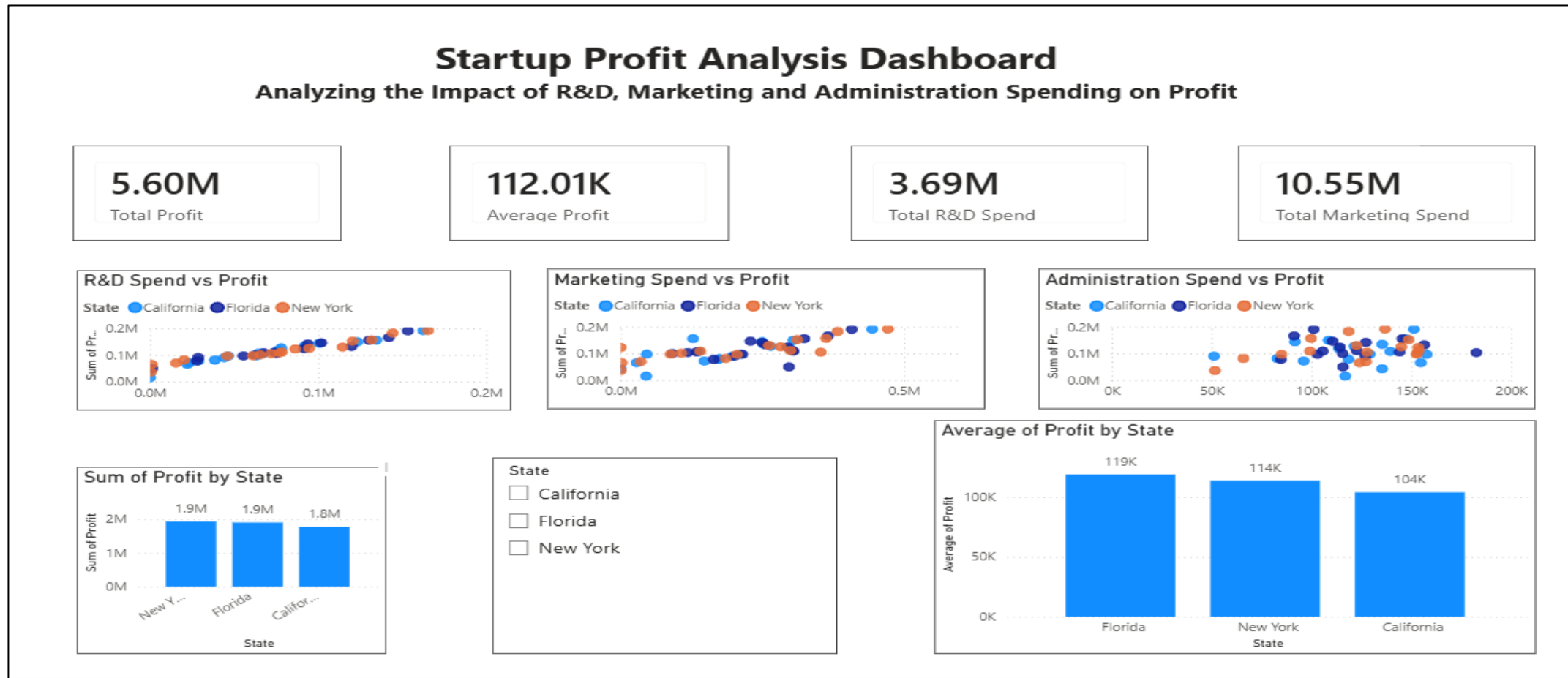
Data Preparation

Data Preparation Activities:

- Imported data from database
- Exported data into CSV format
- Checked for missing values
- Verified data types
- Retained valid zero expenditure values
- Prepared data for visualization and regression analysis

Dashboard Overview

Interactive dashboard created using Power BI to analyze startup profitability.



Regression Analysis

- **Multiple Linear Regression** was used to identify the impact of R&D Spend, Administration Spend, and Marketing Spend on Profit.
- The analysis showed that **R&D Spend** has the strongest positive relationship with Profit.

Regression Results

Excel Regression Results

- $R^2 = 95.07\%$
- Multiple Linear Regression performed using Excel Analysis ToolPak

Python Regression Results

- Implemented using Pandas & Scikit-Learn
- $R^2 = 95.07\%$

Variable	Coefficient
R&D_Spend	0.8057
Administration	-0.0268
Marketing_Spend	0.0272

Excel & Python Implementation

Regression Statistics								
Multiple R	0.97506205							
R Square	0.95074599							
Adjusted R Square	0.94753378							
Standard Error	9232.33484							
Observations	50							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	3	75683964196	2.5228E+10	295.978062	4.52851E-30			
Residual	46	3920856301	85236006.5					
Total	49	79604820497						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	50122.193	6572.352622	7.62621787	1.0574E-09	36892.73332	63351.6527	36892.7333	63351.6527
RD_Spend	0.80571505	0.04514727	17.8463738	2.635E-22	0.714838309	0.89659179	0.71483831	0.89659179
Administration	-0.02681597	0.05102878	-0.52550675	0.60175511	-0.12953157	0.07589964	-0.12953157	0.07589964
Marketing_Spend	0.02722806	0.016451235	1.6550773	0.10471682	-0.00588655	0.06034268	-0.00588655	0.06034268

Excel Regression Output

```
import pandas as pd
from sklearn.linear_model import LinearRegression

df = pd.read_csv("profit_analysis_dataset.csv")

X = df[['RD_Spend', 'Administration', 'Marketing_Spend']]
y = df['Profit']

model = LinearRegression()
model.fit(X, y)

print("Intercept:", model.intercept_)

print("\nCoefficients:")
for feature, coef in zip(X.columns, model.coef_):
    print(feature, "=", coef)

print("\nR Square:", model.score(X, y))

new_data = [
    [21892.92, 81910.77, 164270.7],
    [23940.93, 96489.63, 137001.1]
]

predictions = model.predict(new_data)

print("\nPredicted Profits:")
for i, p in enumerate(predictions, start=1):
    print(f"Record {i}: {p:.2f}")
```

Python Regression Output

```
Intercept: 50122.19298986523

Coefficients:
RD_Spend = 0.8057150499157436
Administration = -0.0268159683947507
Marketing_Spend = 0.02722806480081883

R Square: 0.9507459940683246

Predicted Profits:
Record 1: 70037.90
Record 2: 70554.57
```


Profit Prediction

- $\text{Profit} = 50122.19 + 0.8057(\text{R\&D_Spend}) - 0.0268(\text{Administration}) + 0.0272(\text{Marketing_Spend})$

Record

Record 1

Record 2

Predicted Profit

70,037.90

70,554.57

Observation

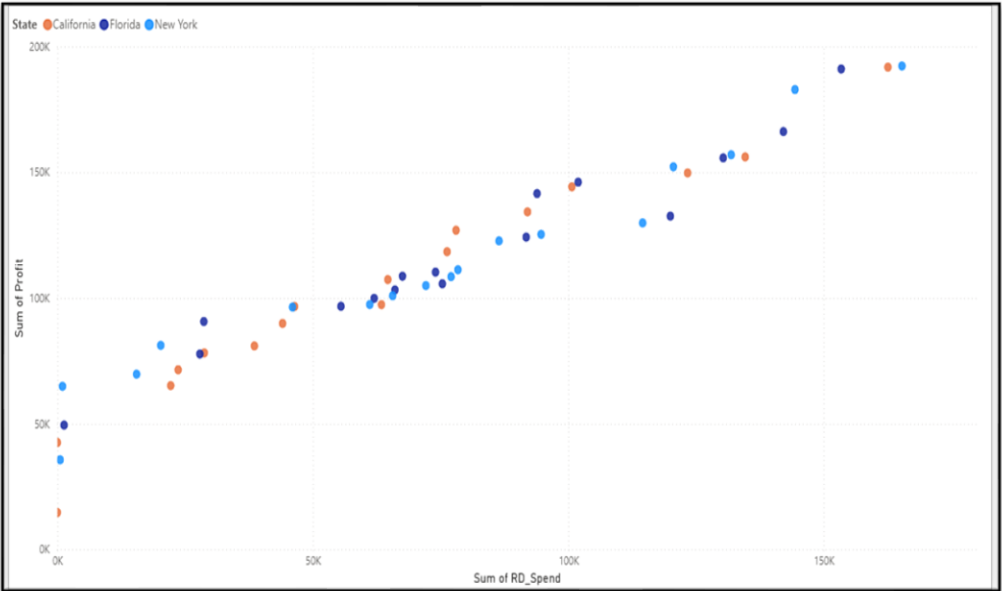
- R&D Spend contributes most to profit growth
- The model achieved 95.07% accuracy

Key Insights

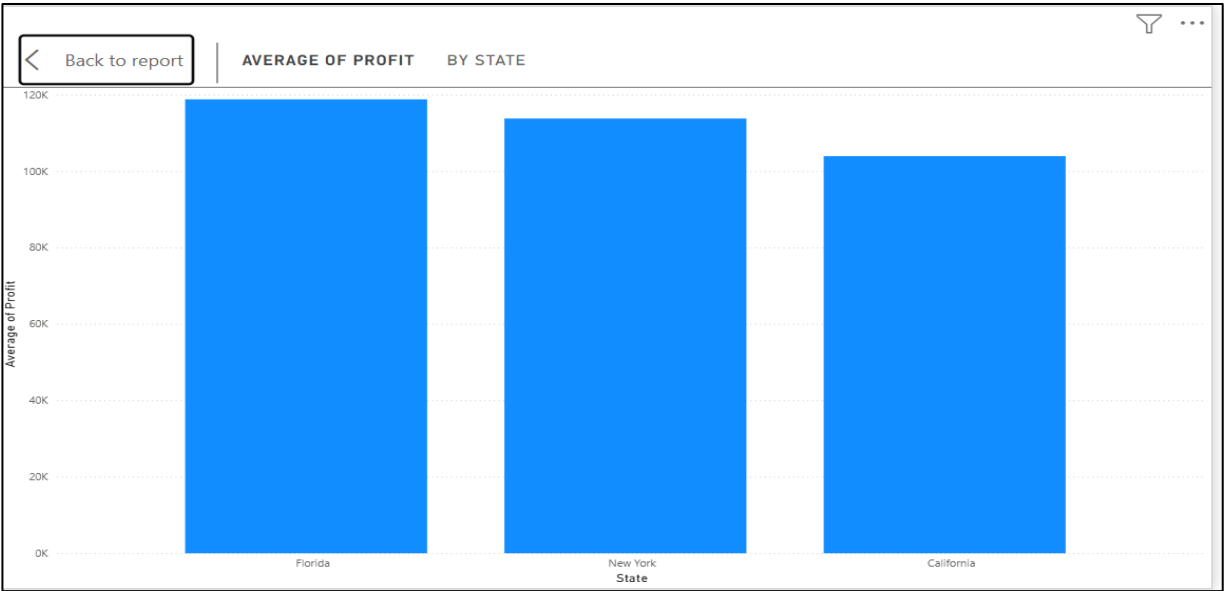
- R&D Spend has the highest impact on Profit
- Marketing Spend has a moderate positive impact
- Administration Spend has relatively low impact on Profit
- Higher investment in R&D generally leads to higher profitability
- Business performance varies across states

Key Visualizations and Findings

Average R&D Spend by State



Profit by State



Recommendations

- Increase investment in R&D activities
- Optimize marketing expenditure for better returns
- Monitor and control administrative expenses
- Use predictive analytics for future business planning
- Continue monitoring investment effectiveness through periodic predictive analysis.

Conclusion

- Regression analysis was used to study the impact of startup expenditures on profit
- R&D Spend was identified as the strongest factor affecting profitability
- Marketing Spend showed a moderate positive influence
- Administration Spend had a relatively low impact on profitability
- The project demonstrates how data analytics can support business decision-making
- The regression model achieved 95.07% accuracy and successfully predicted future profit values
- This project demonstrates practical application of **data analytics, visualization, and machine learning techniques** for business decision-making.